Case Study 27: Advanced Numeric Pattern Generator

1. Problem Statement

Design and implement an algorithmic pattern generator that constructs a dynamic `N × N` concentric square matrix, where layer numbers increment outward from 1 to the perimeter based on minimum distance to grid boundaries.

2. Objectives

- Develop nested iteration logic to dynamically evaluate cell coordinate distances.
- Construct symmetry-preserving multi-layered numeric matrices based on user dimension input.
- Apply boundary distance mathematical formulas for efficient matrix generation.

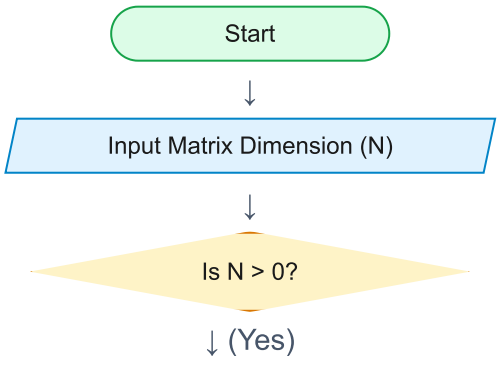
3. Assumptions

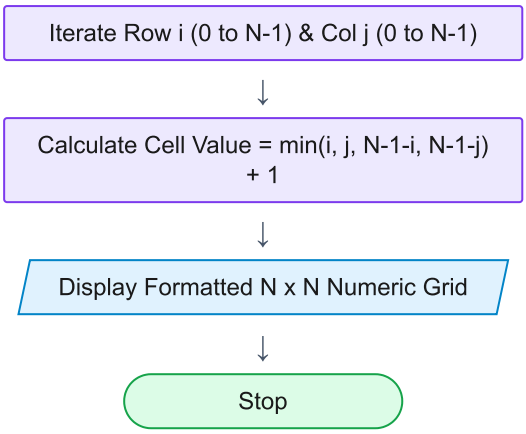
- Input dimension `N` is a positive integer greater than zero.
- Grid indices run from row 0 to `N - 1` and column 0 to `N - 1`.
- Concentric layer values increment from 1 at the outer boundary inward towards the center.

4. Algorithm

1. Start.
2. Read matrix size `N` from user input.
3. Validate that `N > 0`.
4. For each row index `i` from 0 to `N - 1`:
   - For each column index `j` from 0 to `N - 1`:
     - Compute minimum distance to nearest boundary: `dist = min(i, j, N - 1 - i, N - 1 - j)`.
     - Assign cell value: `val = dist + 1`.
     - Append `val` to the current row output buffer.
   - Print the formatted row line.
5. Stop.

5. Flowchart





6. Pseudocode

```
START INPUT N IF N <= 0 THEN PRINT "Invalid Dimension" EXIT ENDIF FOR i FROM 0 TO N - 1 DO row_buffer = [] FOR j FROM 0 TO N - 1 DO min_dist = MIN(i, j, N - 1 - i, N - 1 - j) cell_value = min_dist + 1 APPEND cell_value TO row_buffer ENDFOR PRINT row_buffer JOINED WITH SPACES ENDFOR STOP
```

7. Test Case and Expected Output

Test Case 1: Input N = 5

```
Generated Numeric Pattern for N = 5: 1 1 1 1 1 1 2 2 2 1 1 2 3 2 1 1 2 2 2 1 1 1 1 1 1
```

Test Case 2: Input N = 4

```
Generated Numeric Pattern for N = 4: 1 1 1 1 1 2 2 1 1 2 2 1 1 1 1 1
```

8. Conclusion

The Advanced Numeric Pattern Generator successfully constructs symmetrical concentric square grids by evaluating boundary distance formulas. The algorithm operates efficiently across arbitrary positive dimensions without requiring heavy memory allocation.